Prem Arun P

Virudhunagar, Tamil Nadu | +91-7418156137 prem-arun-p
premarun375@gmail.com premarun37

About

I am a final-year Electronics and Communication Engineering student with a keen interest in the domain of VLSI and digital design. I have an interest in working in hardware design and I have hands-on experience in Verilog and FPGA implementation. I also have working knowledge in SystemVerilog and ASIC Design Flow concepts and I am looking forward to starting my career in ASIC Design where I can use my skills in the best possible way.

Education

Madras Institute Of Technology , Anna University Nov 2022 – Present
BE in Electronics And Communication Engineering- ***CGPA-7.96***

Experience

CodeBind Technology July 2024 – Aug 2024
Student-Embedded System Intern

- Worked on Arduino and ESP32-based projects, gaining hands-on experience in sensor interfacing and serial communication.

Projects

Power Optimization Using Adiabatic Switching for High-Speed Adder 2025
LTSpice

- Designed a 16-bit Weinberger adder using adiabatic switching techniques to significantly cut down dynamic power consumption.
- Implemented PFAL logic paired with an energy-recovering power clock to push the design's efficiency further.

Design and implementation of a 32-bit MIPS single-cycle processor using Verilog 2025
Xilinx Vivado

- Built a MIPS single-cycle CPU from scratch in Verilog, integrating both the datapath and control units into a cohesive design.
- Validated correct instruction execution through hands-on simulation and timing analysis.

Wireless Controlled Pick and Place Robotic Arm 2024
Arduino IDE, Autodesk Fusion 360

- Built a 3-DOF robotic arm controlled wirelessly via Bluetooth using Arduino Uno, enabling smooth real-time object handling.
- Drove servo motors using PWM signals to achieve precise gripping and positional control.

Design and Verification of APB–GPIO Interface Using SystemVerilog 2025
EDA Playground

- Designed an APB–GPIO interface compliant with the AMBA APB protocol, supporting register-level read and write operations.
- Built a structured, object-oriented verification environment in SystemVerilog to thoroughly validate the protocol behaviour.

Automatic Gate Control System for Smart Parking using ESP32 2025
Arduino IDE

- Designed an ESP32-based smart parking system with automatic gate control using sensors and a servo motor.
- Implemented a Binary Search Tree for effective allocation of parking slots with Wi-Fi-based updates.

Skills

Languages: C, Verilog, SystemVerilog, Embedded C
Hardware Platforms: Xilinx Vivado, LTSpice, Linux, OpenLane,Cadence, Proteus, Keil uVision
Soft Skills: Quick Learning, Creativity, Multitasking, Adaptability
Spoken Languages: Tamil (Native), English (Fluent), German (Intermediate), Hindi (Intermediate)

Certifications

- VLSI Design Flow: RTL to GDSII – NPTEL
- Paper Presentation - AISSEWS National Conference (2025)

Position Of Responsibility

- Event Organizer in Symposium
- NSO Volunteer